



interfaceNode - *Datasheet*

Version 1.0.5

Hardware Revision: B.0.x

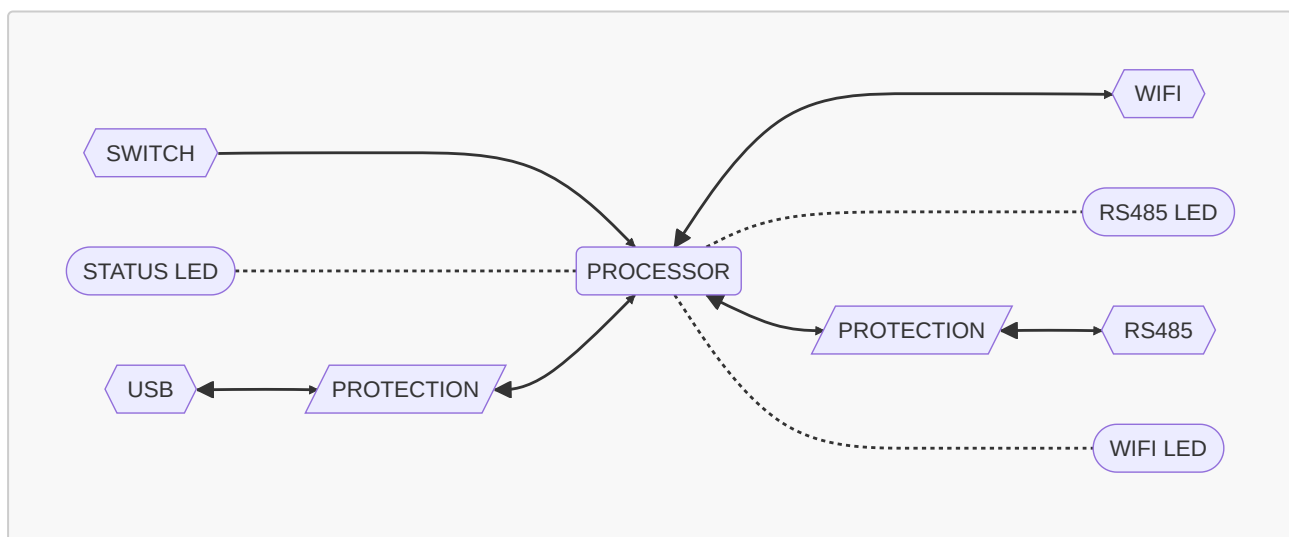
The Atkin Engineering **interfaceNode** is a Wireless RS-485 bridge that provides bridging connections between RS-485, USB, TCP sockets, and WebSockets for various control devices.

front

rear



Block Diagram



Specifications

Mechanical

- Dimensions 120 x 50 x 30 mm (4.72 x 2.36 x 1.18 in)
- Weight 105 grams (2.71 oz)

Electrical

- 12/24 VDC (absolute 5 - 42 VDC) from RJ12 port
- 80 mW average (25 mW - 3 W (peak)) [[^]current]
- Input Protection to ± 42 VDC

RS485

- IEC 61000-4-2 electrostatic discharge (ESD), Level 4 - 8 kV / 15 kV (Contact / Air)
- IEC 61000-4-4 electrical fast transients (EFT), Level 4 - 2 kV
- IEC 61000-4-5 surge immunity, Level 2 - 4 kV
- ± 70 VDC bus fault

USB

- USB serial data via CDC, USB 2.0 full-speed
- Isolation voltage 2.5 kV
- Transient / Surge 5 kV

Services

RS485

- Full / Half duplex [[^]jumpers]
- Termination **None** / **120ohm** [[^]jumpers]
- Protocol aware (DyNet 1 & 2, or raw passthrough)
- Baud rates: 9600 / 19200 / 38400 / 57600 / 115200

WiFi (Access Point / Station)

- IEEE 802.11b/g/n-compliant
- Supports 20 MHz and 40 MHz bandwidth in 2.4 GHz band
- 1T1R mode with data rate up to 150 Mbps
- Center frequency range of operating channel: 2412 ~ 2484 MHz
- WPA2/WPA3 Personal; WPA2/WPA3 Enterprise (802.1X) supported in Station mode
- Station mode falls back to the configuration access point if the network can't be reached after 5 attempts (about two minutes), with the reason recorded in the system log

Bridge

- Bridges data between four interfaces: USB CDC, RS485, TCP sockets (up to 5 clients), and WebSockets (up to 5 clients)
- Protocol aware (DyNet 1 & 2, checksum-validated) or raw passthrough - applies across all four interfaces
- WebSocket traffic can be filtered by DyNet frame type and by opcode, configurable from the web interface
- DyNet discovery beacon (UDP port 9998) and mDNS device discovery - only running services are advertised, with their TLS and login settings

Security

- Unique default password for every device - no shared factory password
- Login on every interface: web interface (session login), WebSocket bridge (one-time token, PDDEG-S style) and TCP bridge (login line); repeated wrong passwords lock the login for an increasing time
- Encryption with TLS 1.2 (ECDHE key exchange, AES-GCM) for the web interface (HTTPS), WebSocket bridge (WSS) and TCP bridge - on by default, each can be turned off for clients that can't use it
- Per-device ECDSA P-256 certificate made on the device; your own certificate can be uploaded instead
- Passwords stored only as salted hashes (PBKDF2-HMAC-SHA256); settings storage (NVS) encrypted
- Full factory reset erases all user data, credentials and the device certificate
- Cross-origin (CORS) access for web apps can be turned off
- Security event log (logins, lockouts, setting changes, firmware updates, factory resets, restart reasons), with an opt-out; no passwords or keys recorded
- Watchdog restarts the device if a data task stops responding

Configuration & Firmware

- Web-based configuration over HTTPS, no software installation required
- USB drive (5 MB) in configuration mode for copying files to and from the device
- System log viewable from the web interface
- Firmware update over WiFi from the web interface, with automatic rollback if the new firmware fails to start

Support and Security Updates

- Security updates are provided free of charge for 7 years after the product's end of life.
- Security fixes released as patch versions (x.y.Z), separately from feature releases, on the [Releases](#) page
- Each release includes a software bill of materials (SBOM) and vulnerability report
- Vulnerability reporting: see [SECURITY.md](#) in the firmware repository

Connections

Front

- USB Type-C, USB 2.0 full-speed port [^{^usbpower}]
- 1x Status LED
- 1x System button

Rear

- 2x RS-485 (parallel RJ12 6P6 ports) [^{^rs485power}]
- 2x RGB status LEDs

Environment

- Operating Temperature: 5° to 40°C ambient (41° to 104°F)
- Storage/Transport Temperature -20° to 65°C ambient (-4° to 149°F)
- Humidity 0 to 90% non-condensing

Compliance

- RF certification: CE, FCC, IC, SRRC, TELEC & BQB [^{^certification}]
- Green certification: REACH/RoHS

Supplied Accessories

- 1x USB Type-C to USB Type-A cable
- 1x RJ12 cable
- 1x Dual RJ12 to pin adapter

© Copyright 2026 [Atkin Engineering](#). All Rights Reserved. [^{^copyright}]

Specifications are subject to change without notice. No representation or warranty as to the accuracy or completeness of the information included herein is given and any liability for any action in reliance thereon is disclaimed.

[^{^current}]: Current requirements dependant on operating modes and environment.

[^{^usbpower}]: USB Type-C port does not supply power.

[^{^rs485power}]: RS-485 (parallel RJ12 6P6 ports) supplies power & data to **interfaceNode**.

[^{^jumpers}]: Termination and duplex are both set via jumpers internal to the enclosure, not configurable from the web interface - changing either requires opening the unit. For duplex specifically: the jumper selects between the receiver always being enabled (full duplex) or tied to the transmit-enable line (half duplex). The firmware drives a single direction-control signal in either case - the jumper, not the firmware, determines which duplex mode results.

[^{^certification}]: Final certification to be completed.

[^{^copyright}]: © Copyright 2026 [Atkin Engineering](#). All Rights Reserved. ABN: 42 715 025 348.